

MOI GIRLS' ELDORET 2025 PREDICTION



POSSIBLE KCSE EXAM COMPUTER STUDIES

451/2

PAPER 2

PRACTICAL

TIME: 2½ HOURS

NAME.....

SCHOOL..... SIGN.....

INDEX NO..... ADM NO.....

(Kenya Certificate of Secondary Education)

INSTRUCTIONS TO CANDIDATES

- a) Indicate your name and index number at the top right hand corner of each printout*
- b) Write your name and index number on the CD/Removable storage media provided*
- c) Write the name and version of software used for each question attempted in the answer sheet provided*
- d) Answer all questions*
- e) All questions carry equal marks*
- f) Passwords should not be used while saving in the CD/Removable storage medium*
- g) All answers must be saved in the CD/Removable storage medium*
- h) Make printout of the answers on the answer sheet*
- i) Arrange your printout and ties/staple them together*
- j) Hand in all printouts and the CD/Removable storage medium used*
- k) This paper consists of four printed pages*
- l) Candidates should check the question paper to ascertain all the pages are printed as indicated and that no question is missing*
- m) Candidates should answer the questions in English*
- n) ALL Data for currency data type should be formatted to Kenya Shillings*
- o) All the system Date and Time should be set to correct Kenya time and Date settings*

QUESTION ONE

a) The data below shows a spreadsheet for loan applicants from a youth fund from a certain county.

MEMBER NO	NAME	DATE OF BIRTH	DATE OF APPLICATION	GENDER	AGE	AMOUNT APPLIED
A001	ABC	1/3/1992	1/1/2016	M		Ksh 200,000.00
B001	CDE	2/6/1999	3/1/2016	F		Ksh 600,000.00
C008	FGH	5/6/1992	3/1/2016	M		Ksh 400,000.00
K001	JKL	2/7/1990	7/1/2016	F		Ksh 700,000.00
S007	MNO	2/9/1960	7/1/2016	M		Ksh 600,000.00
Z006	KRS	4/8/1992	2/2/2016	M		Ksh 500,000.00
A008	TUV	4/8/1960	3/2/2016	M		Ksh 700,000.00
B005	CED	2/6/1995	5/2/2016	F		Ksh 600,000.00
C011	HGF	5/6/1990	10/2/2016	M		Ksh 400,000.00
K012	LKJ	2/7/1970	6/1/2016	M		Ksh 700,000.00
S019	NOM	2/9/1991	4/6/2016	M		Ksh 600,000.00

Type the data as shown in the spreadsheet above name the worksheet as Original save the workbook as application.

(14 Marks)

(b) Copy the worksheet named original above to another worksheet in the same workbook and name it as Evaluation.

(i) Fill the column for the (AGE) which is the difference between DATE OF APPLICATION and DATE OF BIRTH in years to the nearest whole number.

(6 Marks)

(ii) Create a validation in the DATE OF APPLICATION such that it should be after DATE OF BIRTH.

(2 Marks)

(iii) Format all columns having currency Data type to Kenya shilling. **(2 Marks)**

(iv) Create the columns COMMENT and AMT AWARDED.

The COMMENT is based on the following

- If the gender is male and the age by the data of applicable is greater than 30 years the COMMENT is INELIGIBLE otherwise ELIGIBLE.
- Amount awarded if a person is eligible if 80% of the amount applied.

(v) Fill in the two columns for the AMOUNT AWARDED and COMMENT using the formula. If an applicant is INELIGIBLE the amount awarded is left blank. **(10 Marks)**

(c) Copy the worksheet named Evaluation above to another worksheet in the same workbook and name it as final.

(i) Filter out data in the final in the final worksheet leaving out those who have been awarded a loan. **(2 Marks)**

(ii) Create a table showing the total amount awarded to all Males and total awarded to Females and total for both male and females. In the sample table below fill in the values using conditional if statement . **(6 Marks)**

	AMOUNT(Ksh)
MALE	
FEMALE	
<i>Total</i>	

(iii) Draw a pie chart representing the total amount for males awarded and females awarded **(2 Marks)**

(d) Print the following: **(4 Marks)**

- (i) Original worksheet
- (ii) Evaluate worksheet
- (iii) Final worksheet
- (iv) The pie chart

QUESTION TWO

The table below an extract of a manual data structure system used by a librarian of a particular school

ADMN O	S_NAM E	CLAS S	BOOK_N O	TITLE	BORROW TYPE	DATE_BORROW ED	DATE_RETURN ED
123	MIKE	1A	B001	COMPTER STUDIES BK 1	SHORT	2-1-2016	12-1-2016
456	JOHN	1B	A002	COMPREHENSIV E ENG BK 1	LONG	1-2-2016	17-1-2016
789	ADRE W	1C	B003	KIE MATHEMATICS BK 1	LONG	1-3-2016	14-3-2016
987	JAMES	2A	D004	LONGHORN GEOGRAPHY BK2	SHORT	1-4-2016	8-4-2016
654	JACOB	2B	K005	KISWAHILI SHAIRI BK2	SHORT	1-5-2016	9-5-2016
321	NANCY	2C	B006	MATHEMATICS BK2	SHORT	2-3-2016	10-3-2016
879	MARY	3A	C005	HISTORY BK3	LONG	2-4-2016	17-4-2016
564	MERC Y	3B	K009	FOUNDATION CHEMISTRY BK 1	LONG	2-5-2016	14-5-2016
213	PETER	3C	H001	KISWAHILI LUGHA BK 2	LONG	1-4-2016	18-4-2016
123	MIKE	1A	K005	KISWAHILI SHAIRI BK2	SHORT	1-5-2016	19-5-2016
456	JOHN	1B	B006	MATHEMATICS BK2	SHORT	1-4-2016	5-4-2016
789	ADRE W	1C	C005	HISTORY BK3	SHORT	1-4-2016	7-4-2016
987	JAMES	2A	K009	FOUNDATION CHEMISTRY BK 1	LONG	1-5-2016	8-5-2016
654	JACOB	2B	H001	KISWAHILI LUGHA BK 2	LONG	1-4-2016	22-4-2016

The library charges 2 shillings per every book borrowed per day, if the book is not returned in time it attracts a penalty of 5 Shilling per day. The short loan is a maximum of seven days while the long loan is 14 days

a) From the table above create a database called library.

(i) Create THREE tables Student(ADMNO as primary key) and Book(BOOK_NO as primary key) and Borrow(Borrow_id as primary key which is auto number). **(6 Marks)**

(ii) Create relationship between the three tables **(2 Marks)**

(iii) Fill in the data in the three tables **(15 Marks)**

b) Create a the following queries

(i) Query named chargesqry for all charges for books on short loan the query should have the following fields(ADMNO,S_NAME,BOOK_NO,TITLE,BORROW TYPE,DATE_BORROWED,DATE_RETURNED,NO_OF_DAYS_BORROWED,NORMAL_CHARGE,PENALTY_CHARGE,TOTAL_CHARGE)(10 Marks)

(ii) Query named shortchargesqry for all charges for books on short loan the query should have the following fields(ADMNO,S_NAME,BOOK_NO,TITLE,BORROW TYPE,DATE_BORROWED,DATE_RETURNED,NO_OF_DAYS_BORROWED,TOTAL_CHARGE) **(2 Marks)**

(iii) Query named longchargesqry for all charges for books on long loan the query should have the following fields(ADMNO,S_NAME,BOOK_NO,TITLE,BORROW TYPE,DATE_BORROWED,DATE_RETURNED,NO_OF_DAYS_BORROWED,TOTAL_CHARGE) **(2 Marks)**

(iv) Create a report showing the total amount which the library has earned between any two dates on short loan save it as shortbetweendatesrpt format the currency data type to Kenya shillings **(6 Marks)**

(v) Create a report showing the total charged to a student for both long and short loan borrowing in one report save it as studentchrgprt format the currency data type to Kenya shillings **(4 Marks)**

c) Print the following **(4 Marks)**

(i) All tables with data

(ii) Both query with data

(iii) Shortbetweendatesrpt

(iv) studentchrgprt